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#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"

#include "xttcps.h"
#include "xparameters.h"
#include "xscugic.h"
#include "xil_exception.h"

XTtcPs_Config *tC;
XTtcPs tI;

XScuGic_Config *iC;
XScuGic iI;

void eventHdr(){

    XTtcPs_DisableInterrupts(&tI,XTTCPS_IXR_INTERVAL_MASK);

    while(XTtcPs_GetCounterValue(&tI));
    xil_printf("Interrupt\n");

    XTtcPs_ClearInterruptStatus(&tI,XTTCPS_IXR_INTERVAL_MASK);
    XTtcPs_EnableInterrupts(&tI,XTTCPS_IXR_INTERVAL_MASK);
}

void ttcInit(){

int status = 0;
tC = XTtcPs_LookupConfig(XPAR_PS7_TTC_0_DEVICE_ID);
status = XTtcPs_CfgInitialize(&tI,tC,tC->BaseAddress);

if(status == XST_SUCCESS)
    xil_printf("TTC Init Successful\n");
else
    xil_printf("TTC Init Failed\n");

    u32 freq = 1;
    XInterval tmI;
    u8 tmP;

    XTtcPs_Stop(&tI);

    u32 opt = XTtcPs_GetOptions(&tI);
    opt = opt | XTTCPS_OPTION_INTERVAL_MODE | XTTCPS_OPTION_DECREMENT;

    XTtcPs_SetOptions(&tI,opt);

    XTtcPs_CalcIntervalFromFreq(&tI,freq,&tmI,&tmP);

    XTtcPs_SetPrescaler(&tI, tmP);
    XTtcPs_SetInterval(&tI,tmI);

}

void intrInit(){
    int status = 0;
    iC =XScuGic_LookupConfig(XPAR_PS7_SCUGIC_0_DEVICE_ID);
    status = XScuGic_CfgInitialize(&iI, iC, iC->CpuBaseAddress);

    if(status == XST_SUCCESS)
        xil_printf("GIC Init Successful\n");
    else
        xil_printf("GIC Init Failed\n");

    Xil_ExceptionInit();
    Xil_ExceptionRegisterHandler(XIL_EXCEPTION_ID_IRQ_INT, (Xil_ExceptionHandler) XScuGic_InterruptHandler, &iI);
    Xil_ExceptionEnable();

    XScuGic_Connect(&iI, XPAR_XTTCPS_0_INTR, (Xil_InterruptHandler)eventHdr, &tI);
    XScuGic_Enable(&iI, XPAR_XTTCPS_0_INTR);

    XTtcPs_EnableInterrupts(&tI,XTTCPS_IXR_INTERVAL_MASK);

}

int main()
{

    init_platform();
    ttcInit();
    intrInit();

    XTtcPs_Start(&tI);

    while(1);

    cleanup_platform();
    return 0;
}
```